Benchmarking Five Classification Models for a Translate-then-Classify Cyberbullying Detection Pipeline in Code-Mixed Filipino-Bisaya-English Text

Dianna Claire Marie C. Amihan
College of Computer and Information Science
Mapúa Malayan Colleges Mindanao
Davao City, Philippines
dcmAmihan@mcm.edu.ph

John Lawrence S. Alovera
College of Computer and Information Science
Mapúa Malayan Colleges Mindanao
Davao City, Philippines
jlAlovera@mcm.edu.ph

Izzy Mari C. Monteverde
College of Computer and Information Science
Mapúa Malayan Colleges Mindanao
Davao City, Philippines
imMonteverde@mcm.edu.ph

Rhodessa J. Cascaro
College of Computer and Information Science
Mapúa Malayan Colleges Mindanao
Davao City, Philippines
rjCascaro@mcm.edu.ph

Abstract*—Filipino youth increasingly experience cyberbullying in code-mixed Filipino-Bisaya-English text, a form that existing English-only detection systems cannot consistently handle, and no professionally annotated Filipino cyberbullying corpus currently exists to train or evaluate detectors built for this context. A Translate-then-Classify pipeline is developed to address code-mixed Filipino social media text. To evaluate the capability of this approach, five architectures—SVM, mBERT, XLM-RoBERTa Base, XLM-RoBERTa Large, and HateBERT—are benchmarked under uniform preprocessing, translation, and evaluation conditions. The models are evaluated sequentially on a held-out English cross-domain test set (n = 842) and subsequently on a Filipino evaluation corpus (n = 13,724). Because the Filipino corpus's labels were originally assigned for hate speech rather than interpersonal cyberbullying, a licensed psychologist independently reviewed a 150-post sample, and this audit measured how far off hate speech labels are when used as a proxy for cyberbullying ground truth. On the English benchmark, XLM-RoBERTa Large comes out ahead with a Macro-F1 of 0.7692—but that ordering does not carry over after translation, where HateBERT takes the lead on the Filipino set (Macro-F1 = 0.6780) and shows the smallest cross-lingual drop of the five, at 9.2 percent. The audit itself yields a Cohen's kappa of 0.1758, indicating that nearly 22 percent of the corpus's labels don't actually match the cyberbullying construct being studied, so every Filipino-side metric in this paper should be treated as a floor rather than a ceiling. Translation benefits all five models, and the bullying class benefits most, gaining 0.2800 in F1—evidence that translation does more than swap languages; it functions as a form of domain alignment. Fine-tuning the strongest model on expert-labeled Filipino data is the next step this points to.

Keywords—cyberbullying detection, code-mixing, low-resource NLP, cross-lingual transfer, machine translation, transformer models, Filipino, Bisaya

I. Introduction

Among 53 countries measured, the Philippines ranks second in the share of internet users maintaining active social media profiles, with a penetration rate of 99.7 percent [1]—a figure that places social platforms at the core of how Filipino youth communicate day to day. That same connectivity carries a cost. Cyberbullying among Filipino children and adolescents remains a recognized public health concern [2], and exposure to it rises as social media becomes more embedded in everyday life. Harmful content in Philippine digital spaces is rarely confined to a single language. Filipino, Bisaya, and English are routinely blended within the same sentence, sometimes within the same phrase, as part of ordinary online exchange [3], and this mixing is further complicated by non-standard spelling conventions such as Jejemon-style symbol substitution and leetspeak [4]. As a result of these overlapping linguistic conditions, Filipino cyberbullying text falls outside what English-only detection systems are equipped to handle. Monolingual subword tokenizers fragment code-mixed text into semantically empty pieces before classification even begins. Because these models are trained without exposure to Bisaya insults rooted in the local cultural context, they have no basis for recognizing that content as harmful [5], [6].

What stands in the way of building detectors for local languages is less a matter of architecture than a matter of data. Annotated corpora suitable for supervised learning remain severely limited across Philippine languages [7], [8], [9], and, to date, no professionally annotated corpus has been purpose-built for Filipino cyberbullying. The nearest substitutes are hate speech corpora, which capture related but narrower material. Where hate speech targets a person or group specifically because of a protected characteristic, cyberbullying is broader—it covers any repeated, intentional act of online aggression directed at a victim who is not well-positioned to defend themselves [10]. The two constructs do not map cleanly onto one another: they overlap when hostility singles out an identifiable individual but diverge when it is aimed instead at broad political or demographic groups. It follows that a study relying on hate speech labels as a proxy for cyberbullying ground truth is left with a construct mismatch of unknown size, one that can only be surfaced through deliberate auditing.

The Translate-then-Classify architecture addresses this scarcity by routing Filipino input through a machine translation layer before classification, enabling classifiers trained on established English cyberbullying benchmarks to be deployed in the target language without fine-tuning. The existing literature supports the view that cross-domain and cross-lingual adaptation are closely linked problems, and that machine translation can serve as the connecting layer between the two [11], [12], [13], [14]. What remains empirically unsettled, however, is the central design question of which classifier

performs best within such a pipeline once translation is applied to Filipino social media text. It is commonly assumed that classifier rankings established on English benchmarks will carry over—an assumption that tends to favor multilingual models over domain-specific ones—yet this assumption has not been systematically tested in translation, and few studies have isolated the translation layer's contribution through controlled ablation.

This study makes four contributions toward closing these gaps. The first is a Translate-then-Classify pipeline for cyberbullying detection in code-mixed Filipino-Bisaya-English text, built around a seven-stage preprocessing sequence, LLM-based translation, and a token-restoration step. Five classification architectures are then benchmarked on top of this pipeline, all under identical preprocessing, translation, and metric conditions—SVM, mBERT, XLM-RoBERTa Base, XLM-RoBERTa Large, and HateBERT—with each tested against both a held-out English cross-domain set and a separate Filipino evaluation corpus. A licensed psychological professional conducts a label-validity audit next, checking whether the Filipino corpus's original hate-speech labels actually serve as cyberbullying ground truth, and this puts a number on the construct mismatch that arises from repurposing an existing corpus for a task it wasn't built for. A controlled ablation study rounds out the contributions, running across all five models to pin down what the translation layer alone adds to performance.

II. Related Work

Research on automatic detection of online harm in Philippine languages is constrained first by resource scarcity. In their widely used taxonomy of languages by data availability, Joshi et al. [7] place Cebuano among the Rising Star languages— those with a substantial unlabeled web presence but insufficient expert-annotated, task-specific corpora—and surveys of low-resource hate speech detection point to this same absence of task-specific annotated data as the field's dominant bottleneck [8], [15]. Some Philippine-language benchmarking efforts now cover general NLP tasks [9], and local researchers have applied machine learning to Filipino tweets [16] and to multilingual Cebuano-Tagalog-English hate speech [17]; even so, these efforts rest on either small, informally annotated datasets or on hate speech labels whose fit with the psychologically grounded cyberbullying construct has never been verified. A concurrent Southeast Asian benchmark reaches a related conclusion from an independent task and dataset: across Indonesian, Tagalog, Thai, and Vietnamese functional hate-speech tests, Tagalog records the lowest detection accuracy of the four, a gap its authors attribute to linguistic complexity and limited training data, even though their test cases are predominantly monolingual rather than code-mixed [18]. Fortuna and Nunes [10] draw this distinction explicitly: hate speech targets no one in particular, while cyberbullying by definition requires an identifiable victim. No Philippine study to date has had a psychological professional formally audit whether repurposed hate speech labels actually satisfy the cyberbullying construct, which leaves the validity of reported Filipino benchmark metrics an open question.

Transformer architectures for abusive language detection form a second relevant thread. Low-resource targets have benefited from zero-shot and transfer learning enabled by bidirectional transformers [19] and their multilingual variants, mBERT [20] and XLM-RoBERTa [21]. At the same time, HateBERT takes a different approach, specializing in abusive English through retraining on roughly 1.4 million banned Reddit comments [22]. Numbers reported for these models can look impressive when conditions are favorable—F1 near 0.99 has been recorded on curated single-platform data [23]—but that strength does not hold up once a model is tested outside its training distribution, where cross-dataset evaluations show drops exceeding thirty F1 points [24]. In other words, performing well in-distribution on English data says little about how a model will actually behave once deployed. A controlled, side-by-side comparison of classical, multilingual, and domain-adapted architectures on Filipino social media text, run within a single pipeline, is still absent from the literature—leaving open the question of whether multilingual coverage or domain-specific pre-training matters more for this task.

The third thread concerns cross-lingual and translation-based detection. Pamungkas and Patti [11] and Ranasinghe and Zampieri [25] show cross-lingual transfer for abusive language using multilingual embeddings; Casula and Tonelli [12] study machine-translated training data; and Arango et al. [26] apply domain-specific multilingual embeddings to cross-lingual hate speech. Theoretically, this connects to Moreo et al. [13], who argue that cross-domain and cross-lingual adaptation are the same distributional correspondence problem, with translation functioning as the bridge between them. Xian et al. [14] add to this from another angle, finding that zero-shot cross-lingual transfer in multilingual transformers depends mainly on two factors: how well representations align and how much distributional shift separates the source and target. Read together, this work suggests that translating a target language into the source language effectively converts a cross-lingual problem into a cross-domain one. What prior studies have not done, however, is isolate machine translation's contribution through a controlled ablation on the same models and corpus—it is typically treated as just another preprocessing step—and none have checked whether English benchmark rankings actually hold up after translation when applied to code-mixed Filipino text. Recent evidence on large language model translation adds further texture here: LLMs handle code-switched input more robustly than traditional neural MT systems, which depend on aligned parallel data and are brittle to multilingual input, though this advantage narrows for genuinely low-resource language pairs [27]. Whether that robustness extends to preserving hateful or derogatory intent specifically is a separate question; one recent evaluation found translation measurably degrades hate speech detection in code-mixed text, with culturally loaded slang terms often mistranslated into semantically plausible but harmless output [28]. These are precisely the gaps this benchmark addresses.

III. Methodology

A. The ChatSafe Translate-then-Classify Pipeline

This work forms part of a larger undergraduate thesis project developing the ChatSafe cyberbullying detection system, and the present paper focuses specifically on model benchmarking and selection. Fig. 1 presents the end-to-end architecture .

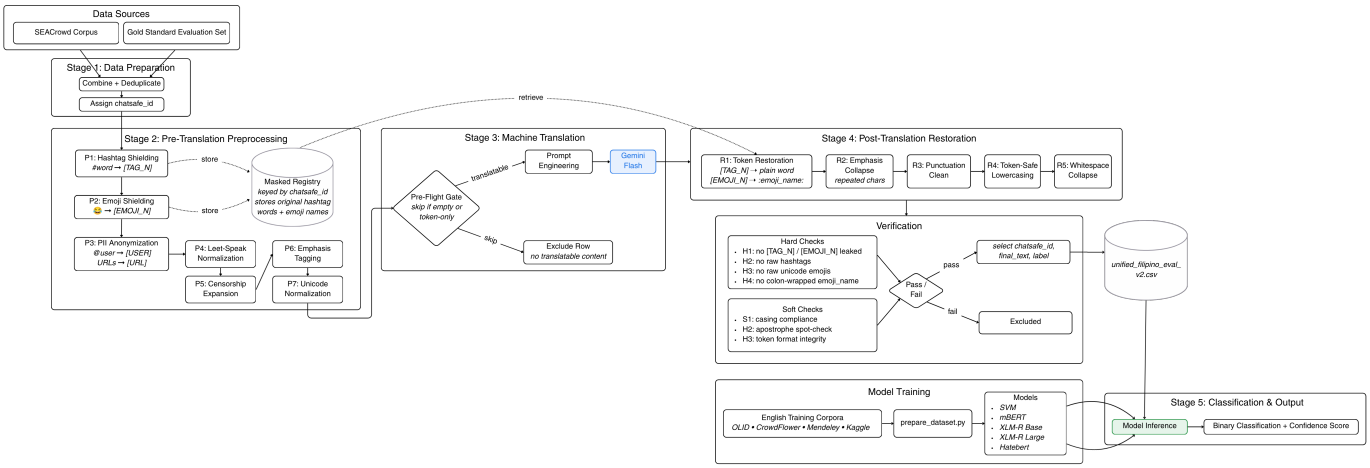


Fig. 1. ChatSafe Translate-then-Classify pipeline architecture.

ChatSafe is a Translate-then-Classify pipeline designed to detect cyberbullying in code-mixed Filipino-Bisaya-English social media text. Instead of classifying code-mixed input directly, the pipeline first shields vulnerable surface features in the raw input, then renders the shielded content into English using generative machine translation. Because of this step, classifiers trained on mature English cyberbullying corpora can operate without any target-language fine-tuning; the system then assigns a binary label (Bullying or Not Bullying) along with a confidence score. The pre-translation stage applies seven sequential preprocessing operations, designed on a shield-and-restore principle, to protect semantic content that translation would otherwise destroy. P1 Hashtag Shield and P2 Emoji Shield mask hashtags and emoji behind stable placeholder tokens. P3 masks personally identifiable information such as user mentions. P4 normalizes leetspeak substitutions, converting adversarial encodings such as b0b0 to their canonical forms. P5 expands self-censored profanity. P6 tags typographic emphasis, such as repeated characters and capitalization, before they are lost in translation. P7 normalizes Unicode variants. Shielded text is submitted to Google Gemini Flash at the translation stage, using a structured prompt that calls for literal, register-preserving translation. Gemini Flash was chosen over dedicated neural machine translation systems and regional language models because it can hold onto culturally embedded derogatory language through contextual pragmatic inference. Systems built on static subword vocabularies break out-of-vocabulary Filipino and Bisaya slurs into subword units that carry no meaning, before translation even begins—a failure documented in NLLB-200 [29]—while SEA-LION, a regional alternative, does not officially support Cebuano and tends to sanitize toxic register [30]. Once translation is complete, five restoration operations follow: shielded tokens are restored, residual emphasis is collapsed, punctuation is cleaned, lowercasing is applied in a way that leaves tokens intact, and whitespace is normalized—yielding the final text that moves on to classification.

B. *Classification Models*

Five classification architectures were selected to span three model families, providing a controlled comparison between classical, multilingual, and domain-adapted approaches. Table I summarizes each model and its role.

TABLE I. CLASSIFICATION MODELS AND ROLES

Model	Family	Pre-training Basis	Role in Benchmark
SVM (TF-IDF)	Classical	None (feature-based)	Non-neural baseline
mBERT	Multilingual transformer	Wikipedia, 104 languages	Multilingual baseline
XLM-RoBERTa Base	Multilingual transformer	CommonCrawl, 100 languages	Scaled multilingual
XLM-RoBERTa Large	Multilingual transformer	CommonCrawl, 100 languages	Largest multilingual
HateBERT	Domain-adapted transformer	BERT retrained on 1.4M banned Reddit comments	Abusive-English specialist

The SVM operates on TF-IDF features and serves as the classical baseline against which transformer gains are measured. mBERT [20] and the two XLM-RoBERTa variants [21] represent the multilingual transfer paradigm at increasing scale. HateBERT [22] represents domain specialization, retrained on the RAL-E corpus of banned Reddit comments and therefore deeply exposed to the colloquial, hostile, and socially aggressive English of the kind that translated cyberbullying content resembles.

C. *Datasets and Label Mapping*

Four publicly available English datasets were integrated into a unified training corpus, selected for documented annotation quality and coverage of distinct abusive language phenomena. Rather than training on a single isolated dataset, a multi-source aggregation strategy was employed to expose the models to a wider distribution of linguistic abuse patterns.

Fig. 2 illustrates the dataset curation and data flow pipeline. Labels across the four sources were harmonized into a binary scheme: Bullying for offensive, abusive, or harassing content, and Not Bullying for neutral content. Following source-level cleaning and SHA-256 hash-based deduplication with cross-split leakage checks, the unified corpus was condensed to 213,090 unique instances (113,907 Not Bullying, 99,182 Bullying), yielding a 1.15-to-1 ratio.

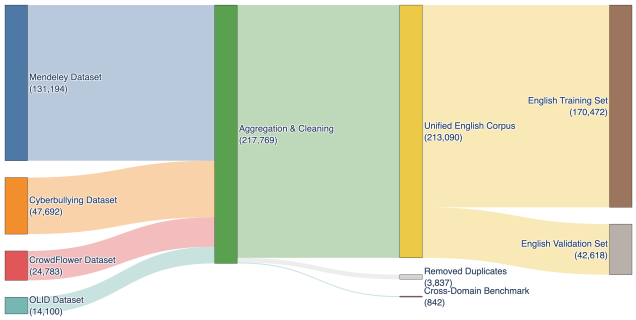


Fig. 2. Data flow for the aggregation, deduplication, and stratified splitting of the multi-source English Training Corpus.

As shown in the flow distribution (Fig. 2), an 80/20 stratified split yields 170,472 training and 42,618 validation instances. The OLID Level A test split (n = 842; 607 Not Bullying, 235 Bullying) is withheld entirely from the unified training pool and serves as a strictly held-out cross-domain English benchmark. The 2.58-to-1 class imbalance motivates the use of Macro-F1 as the primary performance metric. Table II summarizes the sources.

TABLE II. ENGLISH TRAINING DATASET SOURCES

Dataset	Source	Original Size	Role
Cyberbullying Classification Dataset [31]	Kaggle / Twitter	47,692 tweets (6 classes)	Training
OLID [32]	SemEval 2019 Task 6	13,240 train + 860 test	Training + cross-domain test
Mendeley Cyberbullying Dataset [33]	Mendeley Data	131,194 messages	Training
CrowdFlower Hate Speech Dataset [34]	CrowdFlower / data.world	24,783 tweets	Training

The Filipino evaluation corpus was originally compiled by Urbano et al. [35] and released via the authors' public repository, which is cataloged in the SEACrowd data hub [36]. The corpus consists of Filipino social media posts annotated with binary Hate and Non-Hate labels. The dataset's training and test splits were combined and, after deduplication and removal of rows for which translation failed, 13,724 usable instances were retained (6,492 Bullying, 7,232 Not Bullying). Because no professionally annotated Filipino cyberbullying corpus exists, the corpus's hate speech labels were mapped to the cyberbullying label scheme, with Hate treated as Bullying, on the basis that hate speech directed at identifiable individuals or groups in online contexts constitutes a severe form of cyberbullying, consistent with the documented construct overlap in the hate speech literature [10]. This mapping is a proxy assumption rather than a definitional equivalence, and its validity is examined empirically through the label-validity audit described in Section III-D. The corpus is reserved exclusively for evaluation and is never used in training.

D. Label-Validity Audit

To quantify the construct mismatch introduced by the proxy mapping, a sample-based audit was designed, since full professional re-annotation of all 13,724 corpus instances is infeasible. A stratified sample of 150 posts drawn from the corpus was independently assessed by a licensed psychological professional against the operational definition of interpersonal cyberbullying, without access to model predictions or corpus label distributions. Agreement between the corpus labels and the expert assessment is measured with Cohen's kappa,

$$\kappa = \frac{P_0 - P_e}{1 - P_e} \tag{1}$$

where P_0 is the observed agreement, and P_e is the expected chance agreement observed. Kappa is interpreted on the Landis and Koch scale [37], [38]. The audit is designed as a deliberate test of the proxy assumption stated in Section III-C. If agreement falls below the Moderate band, the corpus labels diverge substantially from the cyberbullying construct, and all metrics computed against them must be interpreted as lower bounds on the true model's capability rather than ceilings.

E. Evaluation Protocol and Metrics

All five models are evaluated under identical conditions on two benchmarks, the OLID cross-domain English test set and the translated Filipino evaluation corpus. Classification performance is measured from the confusion matrix components, true positives TP, true negatives TN, false positives FP, and false negatives FN, as

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

The primary ranking metric is Macro-Average F1,

$$Macro{-}F_1 = \frac{1}{C} \sum_{c=1}^{C} F1_c \tag{5}$$

computed as the unweighted mean of per-class F1 across the C=2 classes, chosen because class imbalance in both benchmarks renders accuracy misleading. Cross-lingual degradation for each model is quantified as the relative drop,

$$\Delta = \frac{F_{1_{en}} - F_{1_{fil}}}{F_{1_{en}}} \times 100\% \tag{6}$$

where $F_{1_{en}}$ and $F_{1_{fil}}$ denote Macro-F_1 on the English and Filipino benchmarks, respectively.

To isolate the contribution of the translation layer, a controlled ablation evaluates each model under two conditions on the Filipino corpus: with Gemini Flash translation applied and without translation, applied to raw, preprocessed Filipino text. Model weights, corpus, labels, and metrics are held constant across conditions, so any performance difference is attributable solely to translation.

F. Model Training and Fine-Tuning Hyperparameters

All four transformer models were fine-tuned on the unified English training corpus under a consistent protocol, and the SVM was trained on TF-IDF features of the same corpus. Table

III reports the complete hyperparameter configuration. All stochastic operations used a fixed random seed of 42. Experiments were conducted on Google Colab using an NVIDIA T4 GPU (16 GB VRAM) with PyTorch and Hugging Face Transformers, Scikit-learn for SVMs, and the Google Generative AI SDK for translation.

TABLE III. TRAINING HYPERPARAMETERS

Parameter	HateBERT	XLM-R Base / mBERT	XLM-R Large
Learning rate	2e-5	2e-5	1e-5
Batch size	32	16	16
Epochs	3	3	3
Max sequence length	128	128	128
Optimizer	AdamW	AdamW	AdamW
Warmup ratio	0.1	0.1	0.1
Gradient clipping	1.0	1.0	1.0
Weight decay	0.01	0.01	0.01
Loss function	CrossEntropy	CrossEntropy	CrossEntropy
Validation split	80/20 stratified	80/20 stratified	80/20 stratified
Random seed	42	42	42

IV. RESULTS

A. Label Validity Audit

The expert assessment disagreed with the corpus label on 33 of 150 sampled posts, yielding observed agreement of 78.0 percent and a Cohen's kappa of 0.1758, in the Slight band of the Landis and Koch scale. The disagreement pattern is strongly asymmetric. In 32 cases, the corpus labels a post as Bullying, while the expert assesses it as not constituting interpersonal cyberbullying, compared with a single case in the reverse direction. Inspection of the 32 cases shows that they are dominated by political commentary and by hostile speech directed at electoral candidates, parties, and constituencies rather than at identifiable private individuals. When the corpus's hate speech labels are treated as cyberbullying ground truth, political content is therefore systematically over-represented in the Bullying class. Extrapolated to the full corpus, approximately 1,443 of the 6,492 Bullying-labeled rows (22 percent) diverge from the interpersonal cyberbullying construct. This result confirms the construct mismatch anticipated by the proxy mapping in Section III-C and establishes the interpretive rule applied to every Filipino metric that follows. All Filipino corpus results in this paper are lower bounds on true model capability, not ceilings.

B. English Cross-Domain Benchmark

Table IV presents full metrics for all five models on the OLID cross-domain test set.

TABLE IV. ENGLISH CROSS-DOMAIN BENCHMARK (N=842, OLID TEST SET)

Model	Accuracy	Precision	Recall	Macro-F1	Bullying F1	Not Bullying F1
SVM	76.25%	0.7044	0.7023	0.7033	0.5708	0.8358
mBERT	76.72%	0.7145	0.7290	0.7206	0.6064	0.8347
HateBERT	79.33%	0.7436	0.7497	0.7465	0.6375	0.8555
XLM-R Base	80.17%	0.7535	0.7555	0.7545	0.6469	0.8621
XLM-R Large	81.47%	0.7699	0.7685	0.7692	0.6667	0.8717

Fig. 3 presents these metrics across all five models for visual comparison.

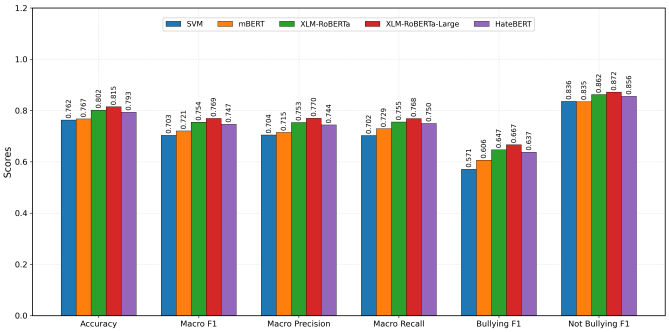


Fig. 3. English cross-domain benchmark, all metrics across five models.

XLM-RoBERTa Large achieves the highest performance across all metrics, and the Macro-F1 gradient from SVM (0.7033) to XLM-RoBERTa Large (0.7692) is monotonic with architectural complexity. All five models exhibit a false-negative-heavy error pattern, missing actual cyberbullying more often than raising false alarms, with false-negative rates on the Bullying class ranging from 43.4 percent for SVM to 33.6 percent for XLM-R Large. HateBERT is the notable exception on the false-positive side, showing the highest false-positive rate among the transformers at 15.2 percent, consistent with its Reddit pre-training, which sensitizes it to colloquial aggressive language. Fifty-three test items are consensus errors misclassified by all five models simultaneously.

C. Filipino Corpus Benchmark

Table V presents full metrics on the translated Filipino evaluation corpus against the original corpus labels, constituting the baseline evaluation.

TABLE V. FILIPINO CORPUS BENCHMARK (N=13,724)

Model	Accuracy	Precision	Recall	Macro-F1	Bullying F1	Not Bullying F1
SVM	63.90%	0.6423	0.6319	0.6289	0.5680	0.6899
mBERT	65.28%	0.6707	0.6421	0.6328	0.5472	0.7185
XLM-R Base	66.76%	0.6770	0.6595	0.6556	0.5914	0.7198
XLM-R Large	68.35%	0.6930	0.6759	0.6732	0.6150	0.7314
HateBERT	68.20%	0.6827	0.6780	0.6780	0.6422	0.7139

Fig. 4 presents these metrics across all five models for visual comparison.

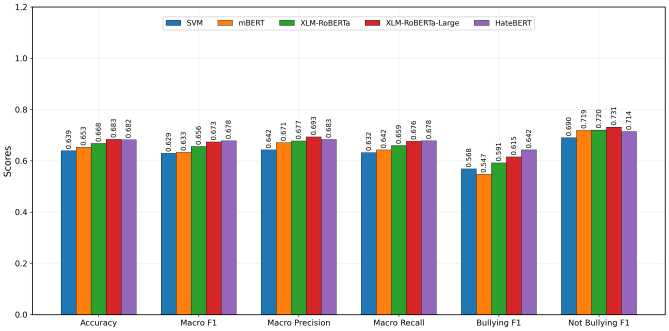
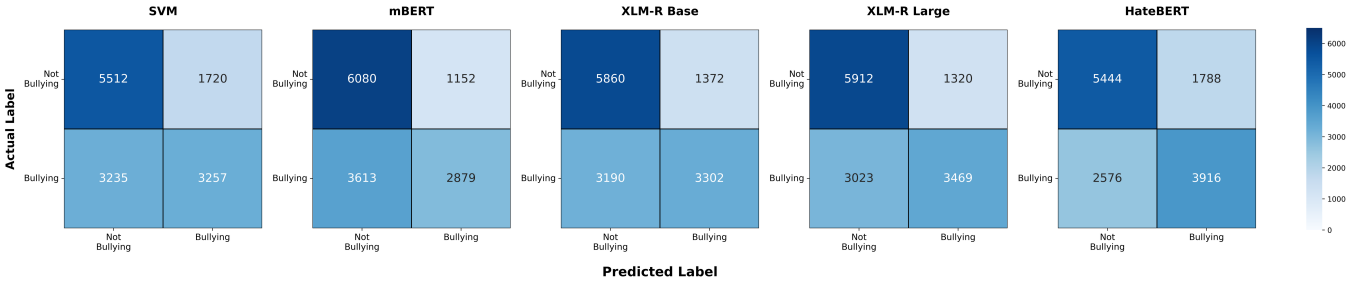


Fig. 4. Filipino evaluation corpus, all metrics across five models.

A ranking reversal is immediately apparent. HateBERT (Macro-F1 = 0.6780) overtakes XLM-RoBERTa Large (0.6732) as the leading model on translated Filipino text, despite ranking lower on the English benchmark. HateBERT also achieves the



(a) SVM (b) mBERT (c) XLM-R Base (d) XLM-R Large (e) HateBERT

Fig. 5(a)-(e). Confusion matrices, Filipino corpus (n = 13,724).

highest Bullying F1 (0.6422) and the lowest false-negative rate (39.7 percent) among all five models, while maintaining the highest false-positive rate (24.7 percent). The false-negative-heavy pattern observed in English is even more pronounced in Filipino across all models, with miss rates for the Bullying class ranging from 39.7 to 55.7 percent. For a safety-oriented detection system in which a missed cyberbullying case is more consequential than a false alarm dismissible by a human reviewer, the trade-off exhibited by HateBERT, more false alarms in exchange for fewer misses, is the preferable error profile. Fig. 5 presents the confusion matrix for each of the five models on the Filipino corpus, isolating the true positive, false negative, false positive, and true negative counts that underlie this trade-off.

The label-validity audit in Section IV-A identified 33 posts out of the 150 sampled in which the corpus's hate speech label diverged from the expert's cyberbullying assessment. To confirm that this construct mismatch measurably suppresses the metrics reported in Table V, those same 33 rows were realigned to the expert assessment, and all five models re-evaluated on the corrected sample, constituting the corrected evaluation. Table VI reports the comparison.

TABLE VI. BASELINE VS. CORRECTED EVALUATION (28-CONSTRUCT DIVERGENT ROWS REALIGNED)

Model	Macro-F1 (Original)	Macro-F1 (Corrected)	Δ F1
SVM	0.6289	0.6302	+0.0013
mBERT	0.6328	0.6337	+0.0009
XLM-R Base	0.6556	0.6564	+0.0008
XLM-R Large	0.6732	0.6738	+0.0006
HateBERT	0.6780	0.6789	+0.0009

Every model improves when the divergent labels are realigned to the cyberbullying construct. The magnitudes are small because only 28 of the estimated 1,443 divergent rows were realigned, but the direction is unanimous, confirming that the proxy mapping measurably suppresses reported performance for every architecture. Extrapolating the per-row effect to full expert re-annotation projects a Macro-F1 recovery of 0.04 to 0.08 points per model, placing expected true performance in the 0.72 to 0.78 range.

D. Ablation of the Translation Layer

Table VII presents an ablation study comparing each model with and without translation on the Filipino corpus.

TABLE VII. ABLATION: WITH VS. WITHOUT TRANSLATION (N=13,724)

Model	Macro-F1 Raw	Macro-F1 Translated	Gain	Bullying F1 Raw	Bullying F1 Translated	Bullying Gain
SVM	0.4614	0.6289	+0.1675	0.2695	0.5680	+0.2985
mBERT	0.5068	0.6328	+0.1260	0.3183	0.5472	+0.2289
XLM-R Base	0.5491	0.6556	+0.1065	0.4024	0.5914	+0.1890
XLM-R Large	0.6171	0.6732	+0.0561	0.5288	0.6150	+0.0862
HateBERT	0.5182	0.6780	+0.1598	0.3622	0.6422	+0.2800

Fig. 6 presents these metrics across all five models for visual comparison.

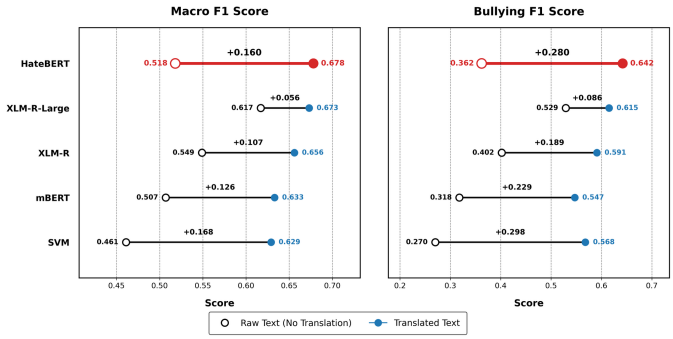


Fig. 6. Impact of translation on Macro-F1 and Bullying F1 across five models.

Translation improves Macro-F1 for all five models, with gains ranging from +0.0561 for XLM-RoBERTa Large to +0.1675 for the SVM, and Bullying F1 gains are systematically larger than Not Bullying gains for every model, indicating that translation disproportionately recovers detection of the harmful class. The pre-translation column reveals the mechanism behind the ranking reversal. Without translation, XLM-RoBERTa Large substantially outperforms HateBERT (0.6171 against 0.5182) because its multilingual pretraining retains partial comprehension of raw Filipino, whereas the English-only HateBERT approaches chance. With translation applied, HateBERT gains +0.1598 in Macro-F1 and +0.2800 in Bullying F1 against XLM-RoBERTa Large's +0.0561 and +0.0862, overturning the ordering. The translation layer is therefore load-bearing for the pipeline, and its contribution differs sharply by model family.

E. Best Model Determination

Table VIII brings together the results from both benchmarks, reporting, for each model, the English Macro-F1, Filipino Macro-F1, and the relative cross-lingual drop Δ (eq. 6).

TABLE VIII. CROSS-DATASET PERFORMANCE SUMMARY

Model	English Macro-F1	Filipino Macro-F1	Absolute Drop	Relative Drop Δ
SVM	0.7033	0.6289	−0.0744	−10.6%
mBERT	0.7206	0.6328	−0.0878	−12.2%
XLM-R Base	0.7545	0.6556	−0.0989	−13.1%
XLM-R Large	0.7692	0.6732	−0.0960	−12.5%
HateBERT	0.7465	0.6780	−0.0685	−9.2%

HateBERT achieves the highest Filipino Macro-F1 of any model and the smallest cross-lingual degradation, with a relative drop of 9.2 percent compared to 12.5 percent for the English leader XLM-RoBERTa Large and 13.1 percent for XLM-RoBERTa Base. The selection rule applied is cross-corpus consistency, requiring the strongest post-translation Filipino performance combined with the smallest cross-lingual degradation, with the ablation confirming the largest translation-conditional gains. Under this rule, HateBERT is the unambiguous selection for the ChatSafe pipeline. English benchmark performance is a proxy for capability under favorable conditions, not a reliable predictor of translated Filipino performance.

V. DISCUSSION

The central finding here is a ranking reversal, and it has a clear explanation. Multilingual models come out ahead when classifying raw Filipino text simply because their pre-training already covers the target language. However, that advantage disappears once translation enters the pipeline—what matters then is a model's grasp of aggressive colloquial English, not how many languages it was trained on. Translation, in effect, turns a zero-shot cross-lingual problem into a cross-domain one, which lines up with the theoretical unification of cross-lingual and cross-domain adaptation [13] and with evidence that transfer performance depends on how well source and target representations align [14]. HateBERT's pre-training on banned Reddit comments places translated Filipino cyberbullying content much closer to its native training distribution, and the ablation results bear this out: a Bullying F1 gain of +0.2800 for HateBERT versus only +0.0862 for XLM-RoBERTa Large, under otherwise identical conditions. The practical takeaway is that once a translation layer bridges the language gap, domain relevance matters more than multilingual breadth when choosing a classifier—and a model's standing on English leaderboards is not a reliable predictor of how it will perform here.

The observed 7-to-13-point cross-lingual drops should be interpreted against the compound difficulty the pipeline faces rather than against idealized single-condition benchmarks. Table IX decomposes the expected suppression into five penalty factors estimated from the cross-lingual literature. The combined expectation places attainable Macro-F1 between 0.58 and 0.68, and HateBERT's observed 0.6780 lands at the top of that range, indicating the pipeline performs near the ceiling of the evaluation conditions.

TABLE IX. PENALTY FACTOR DECOMPOSITION

Penalty Factor	Estimated Macro-F1 Drop
Cross-domain platform shift (training vs. evaluation platforms)	−0.03 to −0.05
Translation noise (sarcasm, slang, culturally embedded insults)	−0.03 to −0.05
Code-mixing complexity (Tagalog, Bisaya, English, leetspeak)	−0.02 to −0.05
Label mismatch (~22% divergent ground truth)	−0.03 to −0.08
Task definition gap (hate speech vs. interpersonal cyberbullying)	−0.02 to −0.04
Expected combined range from English baseline	0.58 to 0.68

This result looks even stronger set against prior work. Numbers like 0.99 F1, reported on heavily curated single-platform data [23], tend not to survive once the data distribution shifts—cross-dataset collapses exceeding 30 points have already been documented [24]. The closest Filipino-language comparison comes from a system fine-tuned directly on in-language Filipino, Cebuano, and Tagalog data, which reports 96.1 percent accuracy but still struggles with recall on Tagalog specifically [17]. Against that backdrop, a Macro-F1 of 0.6780 with zero Filipino training data—and while absorbing five real-world penalties at once—counts as a strong showing, and the corrected-label evaluation suggests the actual figure runs even higher.

The label validity audit also offers a methodological lesson that extends beyond this particular study. Using hate speech corpora as a stand-in for cyberbullying ground truth is routine in low-resource NLP. But this audit finds the construct gap can reach 22 percent, concentrated mostly in political commentary that technically satisfies hate speech criteria without the interpersonal targeting that cyberbullying requires [10]. That doesn't render such corpora unusable—it means the metrics drawn from them are floor values, and papers that present those metrics as point estimates end up underselling what their systems can do. Auditing the proxy assumption directly, as this study does, turns an otherwise invisible validity risk into a measurable, reportable quantity.

Three limitations should be kept in mind alongside these findings. One licensed professional reviewed a sample of 150 posts; because a single rater cannot establish inter-rater reliability for the audit itself, the extrapolated 22% mismatch carries genuine sampling uncertainty and should be treated as indicative, not as a precise population estimate. This also bear directly on how far the finding generalizes: a mismatch rate drawn from one sample, one rater, and one snapshot of the corpus may not hold for a different sample, a different rater, a different platform, or the same corpus at later point in time. Every Filipino-side result in this paper should therefore be read as reliable in direction—hate speech labels systematically over-count cyberbullying—but not necessarily in exact magnitude. The benchmark also relies on a single translation engine. Hence, the translation contribution reported here holds only to the extent that the engine maintains register preservation as it did in testing. And although the evaluation corpora are ecologically grounded, they still capture only a limited portion of Philippine online discourse, leaving open the question of how well this generalizes to other platforms and time periods.

VI. CONCLUSION

This paper presented a Translate-then-Classify pipeline for cyberbullying detection in code-mixed Filipino-Bisaya-English

text, benchmarking five classification architectures against both English cross-domain and Filipino evaluation sets. Three findings stand out. First, rankings from the English benchmark do not carry over to translated Filipino text: HateBERT, despite being domain-adapted rather than multilingual, overtakes the English leader XLM-RoBERTa Large and shows the smallest cross-lingual drop of the five, at 9.2 percent. Second, the translation layer proves load-bearing—it improves every model in the controlled ablation and disproportionately lifts the Bullying class, indicating that translation works as a domain alignment mechanism rather than simply a language bridge. Third, a professional label-validity audit of the repurposed hate speech corpus found a construct mismatch of roughly 22 percent, meaning that every Filipino-side metric reported here is a lower bound and that proxy labels should be audited whenever low-resource benchmark studies are used. Based on its consistency across corpora, HateBERT is selected as the pipeline's classifier. Future work should fine-tune this model on expert-labeled Filipino data as such resources become available, test alternative translation engines—including Philippine-context language models as they mature—and extend end-to-end evaluation of the pipeline against expert-validated ground truth.

REFERENCES

[1] Congressional Policy and Budget Research Department, "Philippine Country Profile in the Digital 2024 Report," *Facts & Figures*, no. 21, House of Representatives, Mar. 2024.

[2] UNICEF Philippines, "Online bullying remains prevalent in the Philippines, other countries," *Press Release*, 06 Sept. 2019. [Online]. Available: https://www.unicef.org/philippines/press-releases/online-bullying-remains-prevalent-philippines-other-countries. [Accessed: Dec. 8, 2025].

[3] M. L. S. Bautista, "Tagalog-english code switching as a mode of discourse," *Asia Pacific Education Review*, vol. 5, no. 2, pp. 226–233, Jun. 2004, doi: 10.1007/bf03024960.

[4] A. L. Ilao and A. C. Fajardo, "SENTIPUBLIKO: Sentiment Analysis of Repost Jejemon Messages using Hybrid Approach Algorithm," *IOP Conference Series Materials Science and Engineering*, vol. 938, no. 1, p. 012010, Oct. 2020, doi: 10.1088/1757-899x/938/1/012010.

[5] R. Zhang et al., "Multilingual Large Language Models Are Not (Yet) Code-Switchers," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, Singapore, December 2023, pp. 12567–12582. doi: 10.18653/v1/2023.emnlp-main.774.

[6] S. Santy, A. Srinivasan, and M. Choudhury, "BERTologiCoMix: How does Code-Mixing interact with Multilingual BERT?," *ACL Anthology*, Apr. 01, 2021. https://aclanthology.org/2021.adaptnlp-1.12/

[7] P. Joshi, S. Santy, A. Budhiraja, K. Bali, and M. Choudhury, "The state and fate of linguistic diversity and inclusion in the NLP world," *ACL Anthology*, Jul. 2020, doi: 10.18653/v1/2020.acl-main.560.

[8] S. Das, A. Dutta, K. Roy, A. Mondal, and A. Mukhopadhyay, "A Survey on Automatic Online Hate Speech Detection in Low-Resource Languages," *arXiv (Cornell University)*, Nov. 2024, doi: 10.48550/arxiv.2411.19017.

[9] M. L. Visperas, C. J. Borjal, A. J. M. Adoptante, D. S. R. Abacial, Ma. M. Decano, and E. C. Peramo, "ITANONG-DS : a collection of benchmark datasets for downstream natural language processing tasks on select Philippine languages," ACL Anthology, Dec. 01, 2023. https://aclanthology.org/2023.icnlsp-1.34/

[10] P. Fortuna and S. Nunes, "A Survey on Automatic Detection of Hate Speech in text," *ACM Computing Surveys*, vol. 51, no. 4, pp. 1–30, Jul. 2018, doi: 10.1145/3232676.

[11] E. W. Pamungkas and V. Patti, "Cross-domain and cross-lingual abusive language detection: A hybrid approach with deep learning and a multilingual lexicon," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics: Student Res. Workshop*, Florence, Italy, Jul. 2019, pp. 363–370, doi: 10.18653/v1/P19-2051.

[12] C. Casula and S. Tonelli, "Hate Speech Detection with Machine-Translated Data: The Role of Annotation Scheme, Class Imbalance and Undersampling," *ACL Anthology*, Mar. 01, 2020. https://aclanthology.org/2020.clicit-1.12/

[13] A. M. Fernández, A. Esuli, and F. Sebastiani, "Distributional correspondence indexing for Cross-Lingual and Cross-Domain sentiment classification.," *Journal of Artificial Intelligence Research*, vol. 55, pp. 131–163, Jan. 2016, doi: 10.1613/jair.4762.

[14] R. Xian, H. Ji, and H. Zhao, "Cross-lingual transfer with class-weighted language-invariant representations," in Proc. Int. Conf. Learn. Representations (ICLR), 2022.

[15] T. Mahmud, M. Ptaszynski, J. Eronen, and F. Masui, "Cyberbullying Detection for Low-resource Languages and Dialects: Review of the State of the art," *arXiv (Cornell University)*, Aug. 2023, doi: 10.48550/arxiv.2308.15745.

[16] J. F. Naga and R. Q. Lavilles, "Deciphering Digital Discourse: Detecting cyberbullying patterns in Filipino tweets using machine learning," *CommIT (Communication and Information Technology) Journal*, vol. 18, no. 2, pp. 167–181, Sep. 2024, doi: 10.21512/commit.v18i2.11094.

[17] P. Pacaldo and J. Matias, "Leveraging machine learning models in developing a Web-Based multilingual hate speech detection system for Cebuano, Tagalog, and English on social media," *Advances in Engineering and Information Sciences*, vol. 1, no. 1, pp. 1–12, Jun. 2025, doi: 10.54610/aeis.v1i1.177.

[18] R. C. Ng, A. Kumaresan, Y. Hu, and R. K.-W. Lee, "SEAHateCheck: Functional Tests for Detecting Hate Speech in Low-Resource Languages of Southeast Asia," *arXiv (Cornell University)*, Mar. 2026, doi: 10.48550/arxiv.2603.16070.

[19] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Minneapolis, MN, USA, Jun. 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.

[20] T. Pires, E. Schlinger, and D. Garrette, "How Multilingual is Multilingual BERT?," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, Florence, Italy, July 2019, pp. 4996–5001. doi: 10.18653/v1/P19-1493.

[21] A. Conneau et al., "Unsupervised Cross-lingual Representation Learning at Scale," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Online, Jul. 2020, pp. 8440–8451, doi: 10.18653/v1/2020.acl-main.747.

[22] T. Caselli, V. Basile, J. Mitrović, and M. Granitzer, "HateBERT: Retraining BERT for Abusive Language Detection in English," in *Proceedings of the 5th Workshop on Online Abuse and Harms (WOAH 2021)*, Online, Aug. 2021, pp. 17–25, doi: 10.18653/v1/2021.woah-1.3.

[23] M. Ahmad, M. Waqas, A. Hamza, S. Usman, I. Batyrshin, and G. Sidorov, "UA-HSD-2025: Multi-Lingual Hate Speech Detection from Tweets Using Pre-Trained Transformers," *Computers*, vol. 14, no. 6, p. 239, Jun. 2025, doi: 10.3390/computers14060239.

[24] D. Antypas and J. Camacho-Collados, "Robust Hate Speech Detection in Social Media: A Cross-Dataset Empirical Evaluation," *arXiv (Cornell University)*, Jul. 2023, doi: 10.48550/arxiv.2307.01680.

[25] T. Ranasinghe and M. Zampieri, "Multilingual offensive language identification with cross-lingual embeddings," *arXiv preprint arXiv:2010.05324*, Oct. 2020. [Online]. Available: https://arxiv.org/abs/2010.05324

[26] A. Arango, J. Pérez, and B. Poblete, "Cross-lingual hate speech detection based on multilingual domain-specific word embeddings," *arXiv (Cornell University)*, Apr. 2021, doi: 10.48550/arxiv.2104.14728.

[27] M. Huzaifah, Z. Weihua, N. Chanpaisit, and K. Wu, "Evaluating Code-Switching Translation with Large Language Models," *ACL Anthology*, May 01, 2024. https://aclanthology.org/2024.lrec-main.565/

[28] F. L. Chan, D. Nguyen, and A. Joshi, "'Is hate lost in translation?': Evaluation of Multilingual LGBTQIA+ hate speech detection," *ACL Anthology*, Dec. 01, 2024. https://aclanthology.org/2024.alta-1.11/

[29] NLLB Team *et al.*, "No language left behind: Scaling human-centered machine translation," *arXiv preprint arXiv:2207.04672*, Jul. 2022. [Online]. Available: https://arxiv.org/abs/2207.04672

[30] R. Ng *et al.*, "SEA-LION: Southeast Asian languages in one network," *arXiv (Cornell University)*, Apr. 2025, doi: 10.48550/arxiv.2504.05747.

[31] J. Wang, K. Fu, and C.-T. Lu, "SOSNet: A Graph Convolutional Network Approach to Fine-Grained Cyberbullying Detection," in *2020 IEEE International Conference on Big Data (Big Data)*, 2020, pp. 1699–1708, doi: 10.1109/bigdata50022.2020.9378065.

[32] M. Zampieri, S. Malmasi, P. Nakov, S. Rosenthal, N. Farra, and R. Kumar, "Predicting the Type and Target of Offensive Posts in Social Media," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, Minneapolis, MN, USA, Jun. 2019, pp. 1415–1420, doi: 10.18653/v1/N19-1144.

[33] N. Ejaz, S. Choudhury, and F. Razi, "A Comprehensive Dataset for Automated Cyberbullying Detection," *Mendeley Data*, V2, 2024, doi: 10.17632/wmx9jj2htd.2.

[34] T. Davidson, D. Warmsley, M. Macy, and I. Weber, "Automated hate speech detection and the problem of offensive language," *arXiv (Cornell University)*, Mar. 2017, doi: 10.48550/arxiv.1703.04009.

[35] R. H. Urbano Jr., J. U. Ajero, A. L. Angeles, M. N. H. Quintos, J. M. R. Imperial, and R. L. Rodriguez, "A BERT-based Hate Speech Classifier from Transcribed Online Short-Form Videos," in *Proceedings of the 2021 5th International Conference on e-Society, e-Education and e-Technology (ICSET)*, Aug. 2021, pp. 186–192, doi: 10.1145/3485768.3485806.

[36] H. Lovenia *et al.*, "SEACrowd: a multilingual multimodal data hub and benchmark suite for Southeast Asian languages," *arXiv (Cornell University)*, Jun. 2024, doi: 10.48550/arxiv.2406.10118.

[37] J. Cohen, "A coefficient of agreement for nominal scales," Educational and Psychological Measurement, vol. 20, no. 1, pp. 37–46, Apr. 1960, doi: 10.1177/001316446002000104.

[38] J. R. Landis and G. G. Koch, "The measurement of observer agreement for categorical data," Biometrics, vol. 33, no. 1, pp. 159–174, Mar. 1977, doi: 10.2307/2529310.